Pauli Was Doing Machine Learning

The Exclusion Principle as a Parsimonious Four-Parameter Model

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The Central Claim

Thesis

When Pauli proposed the exclusion principle in 1925 to organise the complex structure of atomic spectra, he was—in the language we now use—doing **machine learning** and **statistical inference**.

Three acts:

1. **Empirical pattern** (1913–1924): experiments reveal that shell capacities, spectral multiplicities, and Zeeman splittings cannot be explained by three quantum numbers
2. **Stoner's consolidation** (1924): X-ray data show every magnetic sublevel holds exactly *two* electrons
3. **Pauli's minimal model** (1925): a fourth, two-valued quantum number plus one binary constraint predicts *all* observed electron capacities with *zero* free parameters

The episode illuminates—even more starkly than the photoelectric effect—the power of a single theoretical postulate to compress an enormous combinatorial problem into a counting exercise.

Background: Building Blocks of Atomic Structure, 1913–1924

What was known (Bohr 1913; Sommerfeld 1916):

- **Principal** quantum number n: shell energy $E_n = -13.6\,\mathrm{eV}/n^2$
- **Azimuthal** quantum number $l \in \{0, \ldots, n-1\}$: orbital angular momentum
- **Magnetic** quantum number $m_l \in \{-l, \ldots, +l\}$: projection on field axis

States per shell from three numbers:

$$\sum^{n-1}(2l+1) = n^2$$

The Core Puzzle

Shell	n	Predicted n^2	Observed $2n^2$
K	1	1	2
L	2	4	8
M	3	9	18
N	4	16	32

Observed capacity is **exactly twice** the predicted capacity.

Why?

Three Failures of the Three-Quantum-Number Scheme

1. Shell closures unexplained

Bohr proposed noble gases = closed shells: He (2), Ne (10), Ar (18), Kr (36). *No dynamical principle explains why a shell closes at $2n^2$, not n^2.*

2. Anomalous Zeeman effect

Most spectral lines split into **more than 3 components** in a magnetic field. Classical theory (Lorentz) predicts exactly 3. Pauli (1924): "*How can one look happy when he is thinking about the anomalous Zeeman effect?*"

3. Half-integer angular momenta

Landé's (1921) empirical g-factor formula requires total angular momentum J to take **half-integer** values—impossible with three integer quantum numbers.

Landé's Empirical Formula (1921)

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$$

Fits all observed Zeeman spacings perfectly.

Observed g-factors for sodium:

$$g(^2S_{1/2}) = 2.00$$
$$g(^2P_{1/2}) = 0.67$$
$$g(^2P_{3/2}) = 1.33$$

Non-integer values; require $g_s = 2$.

Stoner's 1924 Key Finding: The Decisive Empirical Clue

Edmund Stoner used X-ray spectroscopic level data to count electrons per *sublevel* of each shell.

Stoner's observation: in a magnetic field, the m_l sublevels separate. X-ray data show each m_l sublevel holds **two** electrons.

$$\text{capacity}(n, l) = 2 \times (2l + 1)$$

Summing: $\text{capacity}(n) = 2n^2$. ✓

But: why exactly two per m_l sublevel?
Stoner could not answer this.

Stoner (1924) as Precursor Data

Subshell	$2(2l+1)$	Observed
s $(l=0)$	2	2
p $(l=1)$	6	6
d $(l=2)$	10	10
f $(l=3)$	14	14

ML Language

Stoner performed the analogue of Lenard's feature-importance experiment: he showed that the missing factor is **exactly 2** per sublevel—a binary quantity.

Pauli's 1925 Insight: Model Architecture

Pauli's postulate (January 1925):

Augment the state descriptor with a **fourth quantum number** m_s:

$$m_s \in \left\{+\tfrac{1}{2},\ -\tfrac{1}{2}\right\}$$

Pauli called this a *Zweideutigkeit*—a classically non-describable two-valuedness.

Then assert the **exclusion constraint**:

$$\boxed{n_s \in \{0,\ 1\}}$$

No two electrons share the same 4-tuple $(n,\ l,\ m_l,\ m_s)$.

The Prediction (1925)

Each subshell (n, l) has

$$\underbrace{(2l+1)}_{m_l \text{ values}} \times \underbrace{2}_{m_s \text{ values}} = 2(2l+1)$$

maximum electrons.

Shell capacity:

$$\sum_{l=0}^{n-1} 2(2l+1) = 2n^2$$

Written before a physical interpretation of m_s existed!

The Inductive Bias

The four quantum numbers + exclusion constraint restrict the model class from *all possible occupation rules* to a **unique counting function** with zero adjustable parameters.

The Four-Parameter State Space

Each electron's state is a 4-tuple $(n,\ l,\ m_l,\ m_s)$**:**

#	Quantum number	Range	Identifying observation
1	n (principal)	$1, 2, 3, \ldots$	Chemical periodicity; discrete X-ray edges
2	l (azimuthal)	$0, 1, \ldots, n-1$	Fine structure of X-ray levels; Sommerfeld
3	m_l (magnetic)	$-l, \ldots, +l$	Normal Zeeman effect (3 lines)
4	m_s (spin)	$+\frac{1}{2}, -\frac{1}{2}$	Stoner's factor of 2; alkali doublets
Exclusion: $n_s \leq 1$			Noble-gas stability; closed shells

States per shell (three quantum numbers)

States per shell (four quantum numbers)

The Empirical Dataset: Three Classes of Observations

Class 1: Subshell electron capacities (X-ray spectroscopy + noble-gas chemistry)

Subshell	l	Observed e^-
$1s$	0	2
$2s$	0	2
$2p$	1	6
$3s$	0	2
$3p$	1	6
$3d$	2	10
$4s$–$4f$	0–3	2,6,10,14

Class 2: Spectral term multiplicities

- Alkali metals: doublets ($S = \frac{1}{2}$)
- Alkaline earths: singlets and triplets
- Each multiplicity $= 2J + 1$

Class 3: Anomalous Zeeman component counts

Na transition	Observed	Classical
$^2P_{3/2} \to {}^2S_{1/2}$ (D_2)	**6**	3
$^2P_{1/2} \to {}^2S_{1/2}$ (D_1)	**4**	3

D_2: upper state $J = 3/2$ (4 sublevels), lower $J = 1/2$ (2 sublevels), $\Delta m_J \in \{-1, 0, +1\} \Rightarrow$ **6 lines**.

The Anomaly

6 and 4 lines (not 3) are **impossible** with integer J. The four-parameter model gives the correct count deductively.

Goodness of Fit: Zero Residuals

Subshell	l	Formula $2(2l+1)$	Predicted	Observed	Residual
$1s$	0	2(1)	2	2	0
$2s$	0	2(1)	2	2	0
$2p$	1	2(3)	6	6	0
$3s$	0	2(1)	2	2	0
$3p$	1	2(3)	6	6	0
$3d$	2	2(5)	10	10	0
$4s$	0	2(1)	2	2	0
$4p$	1	2(3)	6	6	0
$4d$	2	2(5)	10	10	0
$4f$	3	2(7)	14	14	0
RSS					**0**

$$\chi^2 = \sum_{\text{subshells}} \frac{(\text{Observed} - \text{Predicted})^2}{\text{Predicted}} = 0$$

Compare with Einstein

Einstein's five-point sodium regression: $R^2 \approx 0.99999$, residuals $\leq 2\,\text{mV}$. Pauli: $\chi^2 = 0$. The counts are *derived*, not *fitted*.

Parameter Identification

Which observation pins down which parameter?

Parameter	What it determines	Identifying observation
$n \in \{1, 2, 3, \ldots\}$	Shell index; n subshell types	Chemical periodicity; discrete X-ray absorption edges
$l \in \{0, \ldots, n-1\}$	Subshell; $2l+1$ magnetic levels	Fine structure of X-ray binding energies; Sommerfeld
$m_l \in \{-l, \ldots, +l\}$	Normal Zeeman splitting into $2l+1$ lines	Normal Zeeman effect; integer-spaced triplets
$m_s \in \{\pm\frac{1}{2}\}$	Factor-of-2 per sublevel; doublet spectra	Stoner's factor of 2; alkali doublets; Landé $g_s = 2$
Exclusion $n_s \leq 1$	Total capacity $= 2n^2$	Noble-gas inertness; absence of all electrons collapsing to $1s$

Key asymmetry

Parameters n, l, m_l were already known. Only m_s is new—and it is identified *indirectly*, through its consequences (the factor of 2), not through a direct measurement.

A Multinomial Likelihood Framing

Treat an electron drawn uniformly from shell n as an observation.

Under Pauli's model, the probability it lands in subshell l is

$$p_l^* = \frac{2(2l+1)}{2n^2} = \frac{2l+1}{n^2}, \quad l = 0, 1, \ldots, n-1.$$

χ^2 **test vs. flat null** ($p_l^{\text{null}} = 1/n$):

Shell	Subshell	Pred. vs. null	χ^2
$n=2$	s ($l=0$)	2 vs. 4	1.00
$n=2$	p ($l=1$)	6 vs. 4	1.00
	$n=2$ total		2.00
$n=3$	s ($l=0$)	2 vs. 6	2.67
$n=3$	p ($l=1$)	6 vs. 6	0.00
$n=3$	d ($l=2$)	10 vs. 6	2.67
	$n=3$ total		5.33

Parsimony

The null model of equal subshell occupancy uses $n-1$ free parameters per shell.

Pauli's model uses **zero**.

AIC favours Pauli by $2(n-1)$ units *plus* the likelihood gain from exact fit.

Generalisation

Pauli's formula extends analytically to any n—shells 5, 6, 7 of xenon and the actinides—without refitting. A purely empirical

Subsequent Confirmation: Spin (Uhlenbeck–Goudsmit, 1925)

In September 1925—eight months after Pauli's paper—Uhlenbeck and Goudsmit proposed that the electron has an **intrinsic angular momentum (spin)** $s = \hbar/2$ with projection $m_s = \pm\frac{1}{2}$ and an anomalous gyromagnetic ratio $g_s \approx 2$.

What this gave Pauli's model:

- A *physical object* for the abstract m_s
- An explanation of the doublet structure of alkali spectra via spin–orbit coupling $\mathbf{L} \cdot \mathbf{S}$
- A derivation of the Landé g-factor formula with $g_s = 2$

Pauli was initially **sceptical**: a classically spinning electron the size of a point particle would need super-luminal surface velocity. He came to accept spin only after Dirac (1928).

Timeline

Jan. 1925	Pauli: exclusion principle; m_s as formal device
Sep. 1925	Uhlenbeck–Goudsmit: m_s = intrinsic spin
1926	Fermi–Dirac statistics
1928	Dirac: spin from relativity
1940	Pauli: spin–statistics theorem
1945	Pauli Nobel Prize

Fermi–Dirac Statistics and the Exclusion Principle

In 1926, Fermi and Dirac independently derived the quantum statistics of particles obeying the exclusion principle:

$$\bar{n}(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/k_B T} + 1}$$

The $+1$ in the denominator is the signature of the Exclusion Principle: $0 < \bar{n} < 1$ for all ε.

At $T = 0$: step function. All states below Fermi energy E_F are full; all above are empty.

Immediate applications:

- Fowler (1926): *white dwarf stability*—electron degeneracy pressure resists gravitational collapse
- Sommerfeld (1927): *metallic heat capacity*—at room temperature only electrons within $k_B T$ of E_F can be excited

Bose–Einstein vs. Fermi–Dirac

$$\bar{n}_{\text{BE}} = \frac{1}{e^{(\varepsilon-\mu)/k_B T} - 1} \quad \text{(bosons)}$$

$$\bar{n}_{\text{FD}} = \frac{1}{e^{(\varepsilon-\mu)/k_B T} + 1} \quad \text{(fermions)}$$

The ± 1 encodes the exclusion or accumulation physics.

Transfer to Astrophysics

The same counting rule that governs atoms explains why white dwarf stars do not collapse.

Dirac (1928) and the Spin–Statistics Theorem (1940)

Dirac (1928): relativistic quantum equation for the electron.

- Spin-$\frac{1}{2}$ emerges *automatically* from combining QM with special relativity—not added by hand
- $g_s = 2$ exactly (including corrections from QED: $g_s = 2.00232\ldots$)
- Retroactively *derives* Pauli's fourth quantum number from first principles

Pauli's spin–statistics theorem (1940):
In any relativistic QFT with locality, causality, and positive energy:

- Half-integer spin particles **must** obey the Exclusion Principle
- Integer spin particles must obey Bose–Einstein statistics

From Postulate to Theorem

1925	Exclusion Principle: a *postulate* with no derivation
1928	Spin from Dirac equation: fourth quantum number derived
1940	Spin–statistics theorem: Exclusion Principle is a *theorem*

Nobel Prize 1945

Awarded to Pauli "for the discovery of the Exclusion Principle, also called the Pauli Principle."

The ML-to-Physics Dictionary

Scientific step	**ML name**	**Who / when**
Capacity $= 2n^2$, not n^2; anomalous Zeeman lines	Anomalous training-set residuals	Bohr–Sommerfeld model by 1922
Stoner: factor of 2 per m_l sublevel	Feature-importance / residual diagnosis	Stoner (1924)
Propose $m_s \in \{+\frac{1}{2}, -\frac{1}{2}\}$; exclusion $n_s \leq 1$	Inductive bias + hypothesis class	Pauli (1925)
State space: (n, l, m_l, m_s) lattice	Feature engineering (add 4th feature)	Pauli (1925)
Count states per subshell: $2(2l+1)$	Forward pass of model	Pauli (1925)
$\chi^2 = 0$: all 10 capacity observations fit	Test accuracy / Confirmation	Spectroscopy 1913–1924
Transferable to molecules, nuclei, stars	Zero-shot transfer learning	Fowler (1926); Sommerfeld (1927)
Spin–statistics theorem	Theoretical derivation of architecture	Pauli (1940)

The Hypothesis Class: Where Pauli Differs from a Naive ML System

Pauli's hypothesis class:

$$\mathcal{H} = \{\text{binary occupation on } (n, l, m_l, m_s) \text{ lattice}\}$$

with $n_s \leq 1$.

- **VC dimension** $= 1$: only one model in the class
- Single observation that confirms $2(2l+1)$ suffices theoretically
- Zero adjustable parameters
- Derived from two postulates, not from data

What a Naive ML System Misses

1. Cannot infer from $(n, l) \to$ count data alone that the explanation is a *two-valued* quantum number
2. Cannot connect capacity data to Zeeman multiplicities without the shared latent variable (quantum number structure)
3. Cannot predict Fermi–Dirac statistics or white dwarf stability from the atomic table
4. Cannot generalise to $n = 5, 6, 7$ without Pauli's analytic formula

The Compression Pauli Achieved

$m_s + n_s \leq 1$ collapsed the hypothesis space from all occupation rules on an arbitrarily rich state space to a **zero-parameter counting function**— the most extreme inductive bias possible.

Pauli vs. a Deep Neural Network

Epoch	Data available	Pauli model	Neural-network outcome
Bohr–Sommerfeld (1922)	Qualitative periodicity	Diagnoses: 3 QN give n^2 not $2n^2$	Sees integers; cannot diagnose missing feature
Stoner (1924)	Subshell counts; 10 integers	Identifies binary 4th QN; zero parameters	Could fit a regression on 10 points, but has no basis for $2(2l+1)$ form
Pauli (1925)	As above + spectral doublets	Predicts all observations; zero free parameters	Memorises; cannot extrapolate to $n = 5$ or Zeeman counts
Modern (large corpus)	All elements + spectroscopy	Same formula; no retraining needed	Could learn $2(2l+1)$ numerically; cannot identify it as exclusion principle

The Core Lesson

Pauli's "training set" was **10 integer observations**. His model has **zero free parameters**. A neural network cannot make use of 10 observations—it needs theory.

Zero-Shot Transfer: From Atoms to Stars

The exclusion principle transfers across all fermion systems:

Domain	How Exclusion Principle applies
Atoms	Shell capacities $2n^2$; chemical periodicity
Molecules	Hückel's $4n+2$ rule for aromatic stability
Solids	Fermi surface; low heat capacity of metals
White dwarfs	Electron degeneracy pressure prevents collapse
Neutron stars	Neutron degeneracy pressure prevents collapse
Nuclear physics	Nuclear shell model; magic numbers

ML Language

Pre-training: Exclusion Principle derived from atomic data

Zero-shot transfer: same principle applied to electrons in metals, nuclear protons/neutrons, stellar interiors

Fine-tuning needed: only the energy spectrum ε_i relevant to each domain—the occupation rule transfers unchanged

Compare with Einstein

Einstein's slope h/e: Na to Li, 0.2% agreement.
Pauli's exclusion rule: atoms to stars, *exact*.

Pauli's Epistemic Situation

Pauli proposed the exclusion principle and the fourth quantum number in January 1925. **Then wrote:**

> *"Already in my original paper I stressed the circumstance that I was* **unable to give a logical reason** *for the exclusion principle or to derive it from more general assumptions... To me, the suggestion of Stoner appeared to be an important step forward, since for the first time* **the number 2** *had to be attributed to the electron itself—to its 'two-valuedness.'"*
>
> — Pauli, Nobel Lecture (1946)

Lesson for Statistics

A parsimonious formal model can unify a body of empirical observations **before** the physical mechanism is understood.

Pauli in 1925 did not know what m_s was. In this he resembles:

- Planck (1900): introduced $h\nu$ without knowing what a "quantum" was
- Einstein (1905): proposed light quanta that conflicted with wave theory
- Millikan (1916): confirmed h/e while rejecting the light-quantum model

Comparison with the Photoelectric Episode

Dimension	**Einstein (photoelectric)**	**Pauli (exclusion principle)**
Anomaly	Stopping potential independent of intensity	Shell capacity $2n^2$ instead of n^2
Key precursor	Lenard (1902): $V_0 \propto \nu$ qualitatively	Stoner (1924): factor-of-2 per sublevel
Postulate	$E = h\nu$ (light quantum)	$m_s \in \{\pm\frac{1}{2}\} + n_s \leq 1$
Model type	Continuous regression	Discrete counting
Free parameters	2 ($\beta = h/e$; $\alpha = -W/e + v_c$)	**0**
Fit statistic	$R^2 \approx 0.99999$ (5 points)	$\chi^2 = 0$ (10 integers)
Transfer	Na to Li, 0.2%	Atoms to stars, exact
Interpretation delayed	Light-quantum debated until 1923	m_s unidentified until Uhlenbeck–Goudsmit (1925)
Grounding achieved	QED (1940s)	Dirac (1928); Pauli spin–statistics (1940)
Nobel Prize	1921 (Einstein)	1945 (Pauli)

The Hypothetico-Deductive Loop

Step	Science	ML
Diagnose $2n^2 \neq n^2$	Empirical anomaly	Training residuals
Stoner: factor-of-2 per sub-level	Key feature identified	Feature importance
Propose $m_s \in \{\pm\frac{1}{2}\}$; $n_s \leq 1$	Theoretical hypothesis	Model architecture
State space: (n, l, m_l, m_s)	Deduction	Hypothesis class
Count $2(2l+1)$ per sub-shell	Prediction	Forward pass
$\chi^2 = 0$: all 10 counts exact	Confirmation	Test accuracy
Fermi–Dirac, white dwarfs, nuclei	Universality	Zero-shot transfer

Modern AI for Science

An AI Feynman-type system given the 10-integer table could find the formula $2(2l+1)$ by symbolic search. What it cannot do:

- Explain *why*—the exclusion principle as a law of nature
- Connect the formula to spectral doublets and Zeeman multiplicities
- Transfer the rule to white dwarfs without re-learning

Theory construction—not number fitting—is Pauli's irreducible contribution.

Summary and Conclusions

The Thesis, Revisited

Pauli in 1925 was doing machine learning—but his decisive step was the one no ML system has yet automated: **specifying the hypothesis class** (what a state space looks like and how many electrons it can hold).

Three principles illustrated:

1. **Extreme parsimony.** One postulate—$n_s \leq 1$—organises the entire periodic table, all atomic spectral multiplicities, Fermi–Dirac statistics, and stellar stability. Zero free parameters.
2. **Theory-first identification.** The fourth quantum number was diagnosed from a qualitative discrepancy (factor of 2), not from a new precision measurement. Confirmation followed theory.
3. **Uninterpreted success.** Pauli proposed m_s without knowing its physical meaning. Planck, Einstein, and Pauli each introduced a formal device that organised data parsimoniously, long before a self-consistent interpretation was available.

The Limit of Pure AI

A data-driven system can recover $2(2l+1)$ numerically but cannot know it is the *exclusion principle*. **Physical meaning requires physical theory.**

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